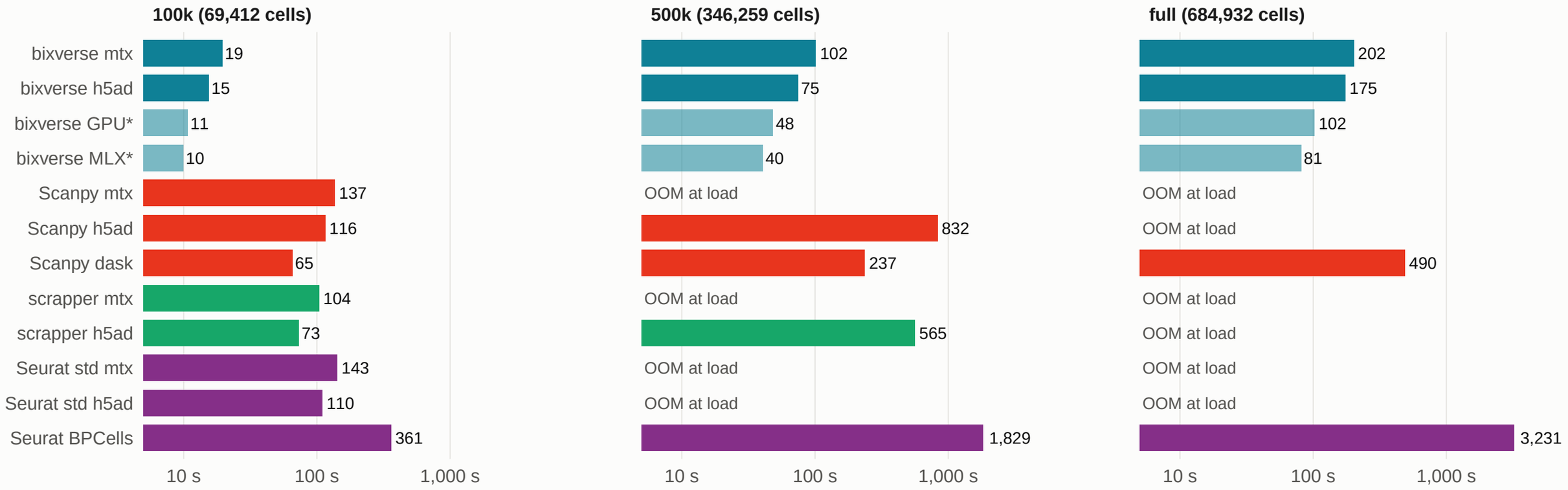
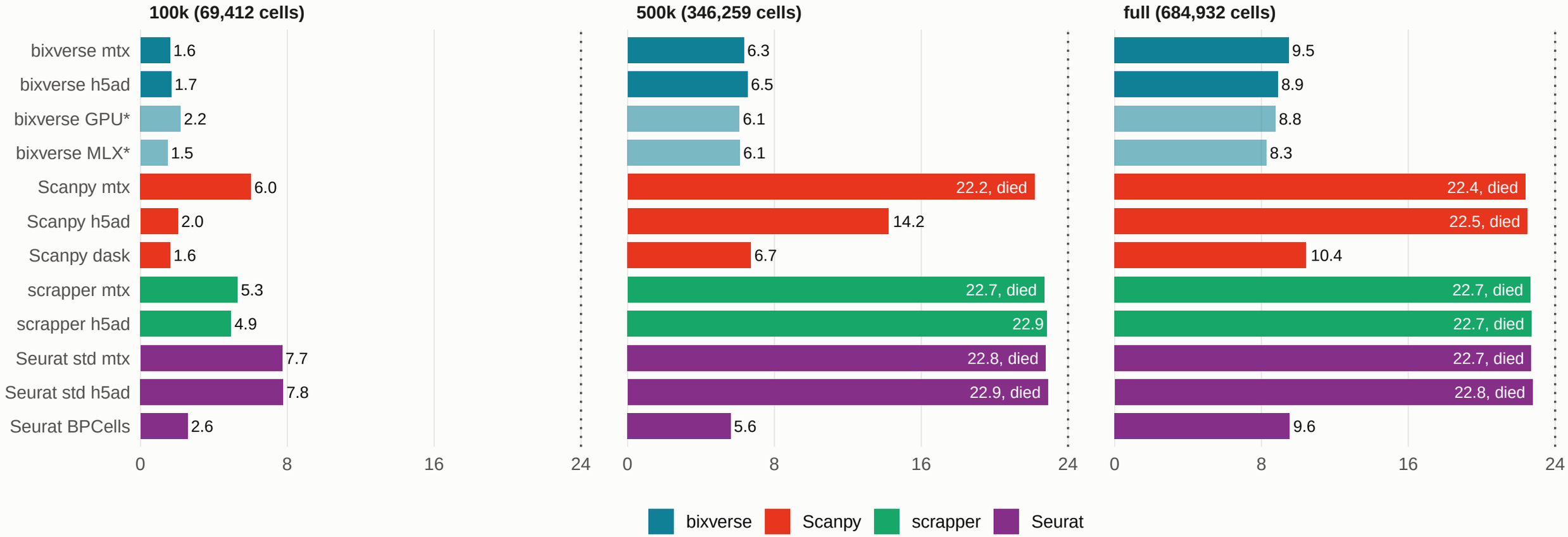
# Parse 1M PBMC: time and memory

Under a 24 GB, 8 core cap. One repeat per run; a run that died shows where.

## Without tSNE




## Peak memory (GB); dotted: the 24 GB cap




Faded: bixverse GPU (wgpu) and MLX, native on Metal without a memory cap; their peak is the macOS memory footprint.
Scanpy dask has no DGE step. A run that timed out in tSNE keeps its time without tSNE.